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DGX Spark, Nemotron3, and NVFP4: Getting to 65+ tps

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DGX Spark, Nemotron3, and NVFP4:
Getting to 65+ tps
42 tps
67+ tps
CUDA graphs
kernel compatibility
container crashes
Accelerated

Running state-of-the-art language models on NVIDIA's DGX Spark isn't as straightforward as `docker run`. After spending considerable time wrestling with CUDA graphs, kernel compatibility issues, and mysterious container crashes, I've documented the complete journey from 42 tokens/second to 67+ tokens/second with full CUDA graph acceleration. Skip to the end for a copy and paste command to have the fastest quantization of the nemotron3 model running.

The Challenge

The DGX Spark features NVIDIA's GB10 "Grace Blackwell" chip — a cutting-edge SoC combining an ARM-based Grace CPU with a Blackwell GPU at compute capability 12.1. This is bleeding-edge hardware, and the software ecosystem is still catching up.

I wanted to run **Nemotron-3-Nano** (specifically the NVFP4 quantized version by cybermotaz) — a hybrid Mamba-Transformer model that promises impressive performance. The model uses:

- **NemotronH architecture**: A hybrid SSM (State Space Model) combining Mamba blocks with Transformer attention
- **NVFP4 quantization:** 4-bit floating point quantization via NVIDIA's modelopt
- **Non-gated activations**: ReLU-squared instead of traditional SiLU/GELU

Attempt #1: Stock vLLM Docker Image

```
docker run --gpus all vllm/vllm-openai:latest \
  serve cybermotaz/nemotron3-nano-nvfp4-w4a16 \
  --quantization modelopt_fp4
```

Result: Complete failure. The stock vLLM image is built for older CUDA architectures and doesn't support SM 12.1 (Blackwell).

Attempt #2: Building vLLM from Source

I built vLLM from the main branch with `TORCH_CUDA_ARCH_LIST="12.1f"`. The `f` suffix enables forward compatibility via PTX JIT compilation.

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New Error:

```
ValueError: NemotronHForCausalLM uses non-gated activation,
which is not supported by FusedMoE
```

The culprit? Nemotron-3-Nano uses **ReLU-squared** activation instead of the gated activations (SiLU) that vLLM's fused MoE kernels expected.

The Non-Gated Activation Fix

Fortunately, PR #29004 added support for non-gated activations. Building from the main branch (post-merge) resolved this issue.

Attempt #3: CUDA Graphs and the SM120 CUTLASS Problem

With non-gated activations working, I tried enabling CUDA graphs for maximum performance:

```
docker run --gpus all my-vllm-image \
  serve cybermotaz/nemotron3-nano-nvfp4-w4a16 \
  --quantization modelopt_fp4 \
  --max-model-len 131072 \
  --gpu-memory-utilization 0.85
```

Result: Container crash during CUDA graph capture with this cryptic error:

```
RuntimeError: Failed to initialize cutlass TMA WS grouped gemm
```

The FlashInfer MoE backend was trying to use CUTLASS kernels that don't fully support SM 12.1 in throughput mode.

The Magic Fix: FlashInfer Backend Mode

After extensive debugging, I discovered the solution:

```
export VLLM_FLASHINFER_MOE_BACKEND=latency
```

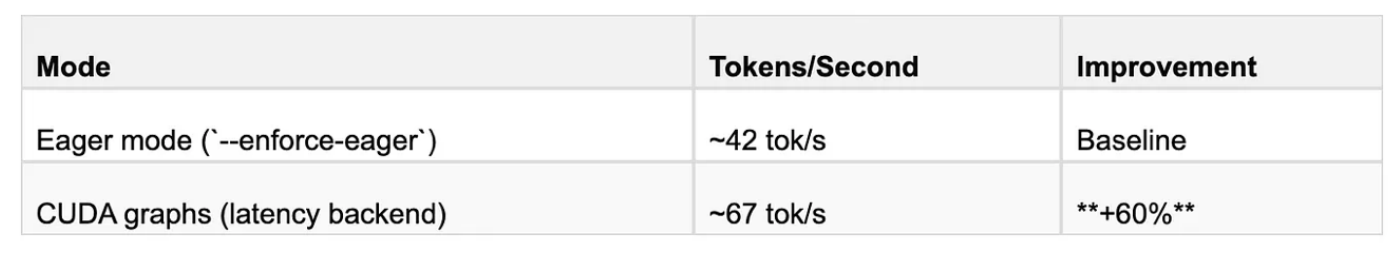
Instead of `throughput` mode (which uses problematic CUTLASS grouped GEMM kernels), `latency` mode uses a different kernel path that's compatible with SM 12.1.

The Final Working Configuration

Here's the one-liner that achieves 65+ tokens/second with full CUDA graph acceleration, including reasoning and tool calling support:

```
docker run --rm -it --gpus all --ipc=host -p 8000:8000 \
  -e VLLM_FLASHINFER_MOE_BACKEND=latency \
  -v ~/.cache/huggingface:/root/.cache/huggingface \
  avarok/vllm-dgx-spark:v11 \
  serve cybermotaz/nemotron3-nano-nvfp4-w4a16 \
  --quantization modelopt_fp4 \
  --kv-cache-dtype fp8 \
  --trust-remote-code \
  --max-model-len 131072 \
  --gpu-memory-utilization 0.85 \
  --enable-auto-tool-choice \
  --tool-call-parser qwen3_coder \
  --reasoning-parser deepseek_r1
```

Performance Results



Mode	Tokens/Second	Improvement
Eager mode (`--enforce-eager`)	~42 tok/s	Baseline
CUDA graphs (latency backend)	~67 tok/s	**+60%**

That's a 60% performance improvement just from enabling CUDA graphs with the correct backend configuration.

Building Your Own Image

The Docker image is available at `avarok/vllm-dgx-spark:v11`, but if you want to build it yourself, here's the key Dockerfile configuration:

```
FROM nvidia/cuda:13.0.2-cudnn-devel-ubuntu24.04
```

```
# Critical: Set architecture for Blackwell GPU
ENV TORCH_CUDA_ARCH_LIST="12.1f"

# Critical: Use latency backend for SM12.1 compatibility
ENV VLLM_FLASHINFER_MOE_BACKEND=latency

# Install PyTorch with CUDA 13.0 support
RUN pip install torch torchvision torchaudio --index-url
https://download.pytorch.org/whl/cu130

# Install FlashInfer for CUDA 13.0
RUN pip install flashinfer-python -i
https://flashinfer.ai/whl/cu130/torch2.7

# Build vLLM from main branch
RUN git clone https://github.com/vllm-project/vllm.git && \
    cd vllm && \
    pip install --no-build-isolation . -v
```

The full Dockerfile and build instructions are available on HuggingFace at Avarok/vllm-dgx-spark.

Key Configuration Parameters Explained

- **`--quantization modelopt_fp4`**: Uses NVIDIA's FP4 quantization format
- **`--kv-cache-dtype fp8`**: FP8 KV cache reduces memory usage while maintaining quality
- **`--max-model-len 131072`**: 128K context window (ideal for the DGX Spark)
- **`--gpu-memory-utilization 0.85`**: Leave headroom for CUDA graph capture
- **`--ipc=host`**: Required for proper shared memory access
- **`VLLM_FLASHINFER_MOE_BACKEND=latency`**: The critical fix for SM12.1 compatibility
- **`--enable-auto-tool-choice`**: Enables automatic function/tool calling for agentic workflows
- **`--tool-call-parser qwen3_coder`**: Parser for structured tool call responses
- **`--reasoning-parser deepseek_r1`**: Parser for reasoning tokens (think tags)

Troubleshooting Common Issues

"Free memory increased during profiling"

Reduce `--gpu-memory-utilization` to 0.75-0.80. This happens when torch.compile releases buffers during memory profiling.

Container crashes during CUDA graph capture

Ensure `VLLM_FLASHINFER_MOE_BACKEND=latency` is set. The throughput backend uses CUTLASS kernels that don't work on SM12.1.

Model loading fails with "non-gated activation" error

You need vLLM from the main branch (after May 2025) which includes PR #29004.

Conclusion

Getting optimal performance on DGX Spark requires understanding the nuances of CUDA 13.0, Blackwell architecture, and vLLM's various backend options. The key discoveries were:

1. **Build for SM 12.1f** with forward compatibility
2. **Use the latency backend** for FlashInfer MoE operations
3. **Leave GPU memory headroom** for CUDA graph capture
4. **Build from vLLM main** for non-gated activation support

With these configurations, the DGX Spark delivers impressive 67 tokens/second inference on Nemotron-3-Nano — making it a capable edge AI platform for production workloads.

. . .

Credits to cybermotaz for the excellent NVFP4 quantization of Nemotron-3-Nano.

Docker image: `avarok/vllm-dgx-spark:v11` *HuggingFace repo: Avarok/vllm-dgx-spark*

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What are your thoughts?

Cédric M.
Jan 5

Thank you for the Docker image; it works perfectly, and I can confirm the 60% performance improvement with Nemotron 3. I've just started experimenting with my Spark, and resources like this one are incredibly helpful—I'd even call it a gem.

I noticed… more

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